BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. Water gas is a mixture of .....

(A) CO and H_2
(B) C and H_2
(C) CO and H_2O
(D) C and H_2O

Q2. Blister copper is refined through .....

(A) concentration
(B) reduction
(C) smelting
(D) electrolysis

Q3. The equilibrium constant expression for the reaction $N_2+3H_2 \rightleftharpoons 2NH_3$ is .....

(A) $[NH_3]^2/([N_2][H_2]^3)$
(B) $[2NH_3]/([N_2][H_2]^3)$
(C) $[N_2][3H_2]/[2NH_3]$
(D) $[2NH_3]/([N_2][3H_2])$

Q4. The coldest region in the atmosphere is .....

(A) troposphere
(B) stratosphere
(C) mesosphere
(D) thermosphere

Q5. Chlorine acts as bleaching agent in the presence of .....

(A) HCl
(B) H_2O
(C) NaOH
(D) HNO_3

Q6. All are the raw materials for the manufacture of urea except .....

(A) CO_2
(B) NH_3
(C) H_2O
(D) NaCl

Q7. Which one of the following decolourises Br_2 water?

(A) ethane (B) ethene

(C) propane (D) methane

Q8. Alkyl halide is composed of carbon, hydrogen and .....

(A) oxygen (B) nitrogen

(C) halogen (D) sulphur

Q9. Peptide linkage is present in .....

(A) carbohydrates (B) lipids

(C) proteins (D) vitamins

Q10. The substance that ionizes completely in an aqueous solution and gives a higher concentration of H^+ ions called .....

(A) weak base (B) strong base

(C) weak acid (D) strong acid

Q11. Dehydration of ethyl alcohol with conc. H_2SO_4 results in the formation of .....

(A) ethane (B) methane

(C) ethyne (D) ethene

Q12. Which one of the following is a basic salt?

(A) KCl (B) NaCl

(C) Pb(OH)Cl (D) $KHSO_4$

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 32

Q1. Attempt any eight of the following short questions. Each question carries 4 marks

(i) Differentiate between reversible and irreversible reactions. Give one example of each.

(ii) Distinguish strong acids from weak acids. Give two examples of each.

(iii) What are cyclo alkanes? Give two examples.

(iv) What happens when alkyl halide is reduced?

(v) Describe the functions of nucleic acid.

(vi) Discuss the composition of atmosphere.

(vii) Enumerate the advantages of hard water.

(viii) Define metallurgy. What are its steps?

(ix) Write down four important uses of urea.

(x) Define pH and pOH.

(xi) How are alkenes prepared by dehydration of alcohols?

SECTION-C

Maximum Marks: 21

Note: Attempt any three of the following questions. Each question carries 7 marks

Q2.

(i) What are salts? Describe the different types of salts.

(ii) Relate the active mass with rate of a chemical reaction.

Q3.

(i) Define functional group. Give some examples of functional groups containing oxygen.

(ii) Explain the halogenation reaction of alkynes.

Q4.

(i) What are pollutants? Differentiate between primary and secondary pollutants.

(ii) How are proteins important to living organisms?

Q5.

(i) What is slaked lime? How is slaked lime produced?

(ii) Increase in concentration of CO_2 causes greenhouse effect, justify it.